CyberNova

Sprint 1 Project Documentation

For

BusNova Bus Management Version 1.0

Submission date: 11 August 2026

For: Mr. D. Boucher

Declaration of Work:

We, CyberNova, declare that:

- The system developed is our original work and has not in part been submitted for the purposes of final assessment for any other 3rd year project module,
- We have the permission to use images shown in our system and have created our own logos that are used in our system,
- Where we have made use of programming assistance, we acknowledge this fact in the acknowledgements section below,
- We did not outsource the building of the entire system to a third party.

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Role Key:
PM: Project Manager *(only one member can be PM at any given time)*
BA: Business Analyst
DO: Documenter
PR: Programmer
A team member can complete multiple roles.

Acknowledgements:

We would like to thank:

1. Lecturers:
    - Mr G. Mainoti
    - Dr E. Chindenga (coding expert)
    - Mr T. Breetzke (coding expert)
    - Prof R. Piderit (business analysis)

2. Our Mentor: Anelisa Gobingca

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System Matrix

Business Process	Reference	Use Case	Table	Screen	Report
Register an account	UC1	Register	User	SC2	List of Registered Users
Login to the system	UC2	Login	User	SC3	Login Activity Report
Register buses	UC3	Register bus	Bus	SC4	Registered Buses Report
Create routes	UC4	Create route	Route	SC5	Route List Report
Assign routes	UC5	Assign route	Route Assignment	SC6	Route Assignment Report
Monitor bus operations	UC6	Monitor operations	Bus	SC8	Bus Operations Report
Review reports on route completion	UC7	View reports	Route	SC9	Route Completion Report
Review fuel efficiency	UC8	Review fuel	FuelLog	SC10	Fuel Efficiency Report
View assigned routes & schedule	UC9	View assigned routes	RouteAssignment	SC7	Driver Schedule Report
Report fuel usage	UC10	Report Fuel	FuelLog	SC11	Fuel Usage Report
Report maintenance issues	UC11	Report Issue	MaintenanceLog	SC12	Maintenance Issues Report
Update trip status	UC12	Update Status	Bus	SC13	Trip Status Report

View bus routes	UC13	View Routes	Route	SC14	Route Report
Check availability	UC14	Check Availability	Bus	SC15	Bus Availability Report
Purchase ticket	UC15	Purchase Ticket	Ticket	SC16	Ticket Sales Report
Receive confirmation email	UC16	Receive Confirmation	Ticket	SC17	Booking Confirmation Report
View maintenance schedule	UC17	View Maintenance Schedule	MaintenanceLog	SC18	Maintenance Schedule Report
Log reports and servicing activities	UC18	Log Service Activities	MaintenanceLog	SC19	Service History
Update vehicle condition	UC19	Update Vehicle Status	MaintenanceLog, Bus	SC20	Vehicle Condition Report
Receive alerts for upcoming and overdue services	UC20	Receive Maintenance Alerts	MaintenanceLog	SC21	Maintenance Alert Report

Table 1: BusNova System Matrix

System Overview

BusNova is an integrated Bus Management System designed to centralise and streamline the management of bus fleet operations, transportation services and vehicle maintenance. The system provides role-based functionality for Fleet Managers, Bus Drivers, Passengers and Maintenance Staff, enabling users to perform activities according to their responsibilities. These include user registration and authentication, bus registration, route creation and assignment, trip-status updates, passenger ticket purchasing, fuel monitoring, maintenance management and operational reporting.

The system connects these processes through a centralised database containing information relating to users, buses, routes, route assignments, tickets, fuel logs and maintenance records. Fleet Managers can allocate available buses and drivers to routes while the system validates availability to prevent scheduling conflicts. Drivers can access assigned routes and schedules, while Maintenance Staff can monitor servicing schedules, record maintenance activities and receive alerts for overdue services. Passengers can view routes, check availability, purchase tickets and receive booking confirmations.

BusNova supports Sustainable Development Goal 9 (Industry, Innovation and Infrastructure) by digitally integrating and improving transportation infrastructure management. It supports SDG 11 (Sustainable Cities and Communities) by facilitating organised and accessible public transportation services. The system also contributes to SDG 12 (Responsible Consumption and Production) through fuel-usage and fuel-efficiency monitoring, as well as preventative vehicle maintenance. These functions collectively promote more efficient resource utilisation and sustainable transport operations.

Overall, BusNova provides a unified platform that improves operational coordination, information accuracy, fleet visibility and decision-making across the bus management environment.

1. Section A: Business Process Description

1.1. Use Case Diagram

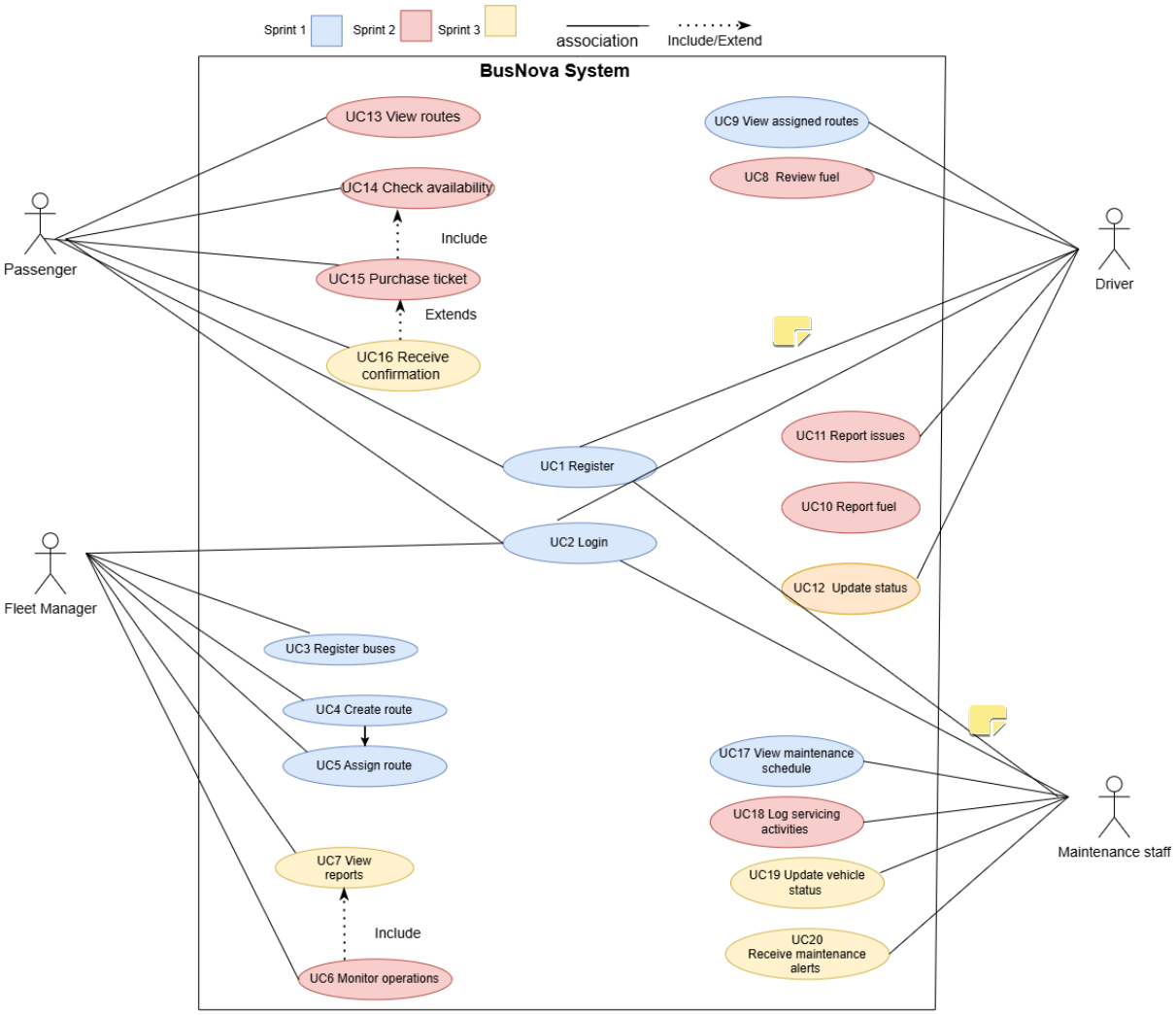


Figure 1: BusNova Use Case Diagram

1.2. Use Case Narrative

1.2.1. Register Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC1**		
User Case Name	**Register**		
Actor(s)	Maintenance Staff, Driver, Passenger		
Brief Description	This use case allows a new user to register an account by entering their personal details. The system validates the information, checks whether the email address already exists, encrypts the password, stores the user's information in the database, creates the account, and redirects the user to the login page upon successful registration.		
Main Flow of Events	Actor	System	
	1. Opens the Registration page. 2. Enters registration details (e.g., name, surname, email, password, confirm password). 3. Clicks the Register button.	1,2. Displays the registration form. 2.1. Validates the entered information. 2.2. Checks whether the email address validation in the database. 3.1. Displays a registration success message if Valid. 3.2. Display a validation error when not valid. 3.3. Redirects the user to the Login page.	
Alternative Flows of Events	If required fields are missing or invalid, the system displays validation error messages and returns the user to the registration form to correct the information.		
Created by	Elihle Nteyi	Last Modified by	Yonwaba Mqulwana
Date	01 August 2026	Date	05 August 2026

Table 2: Register Use Case Narrative

Activity Diagram 1: Register

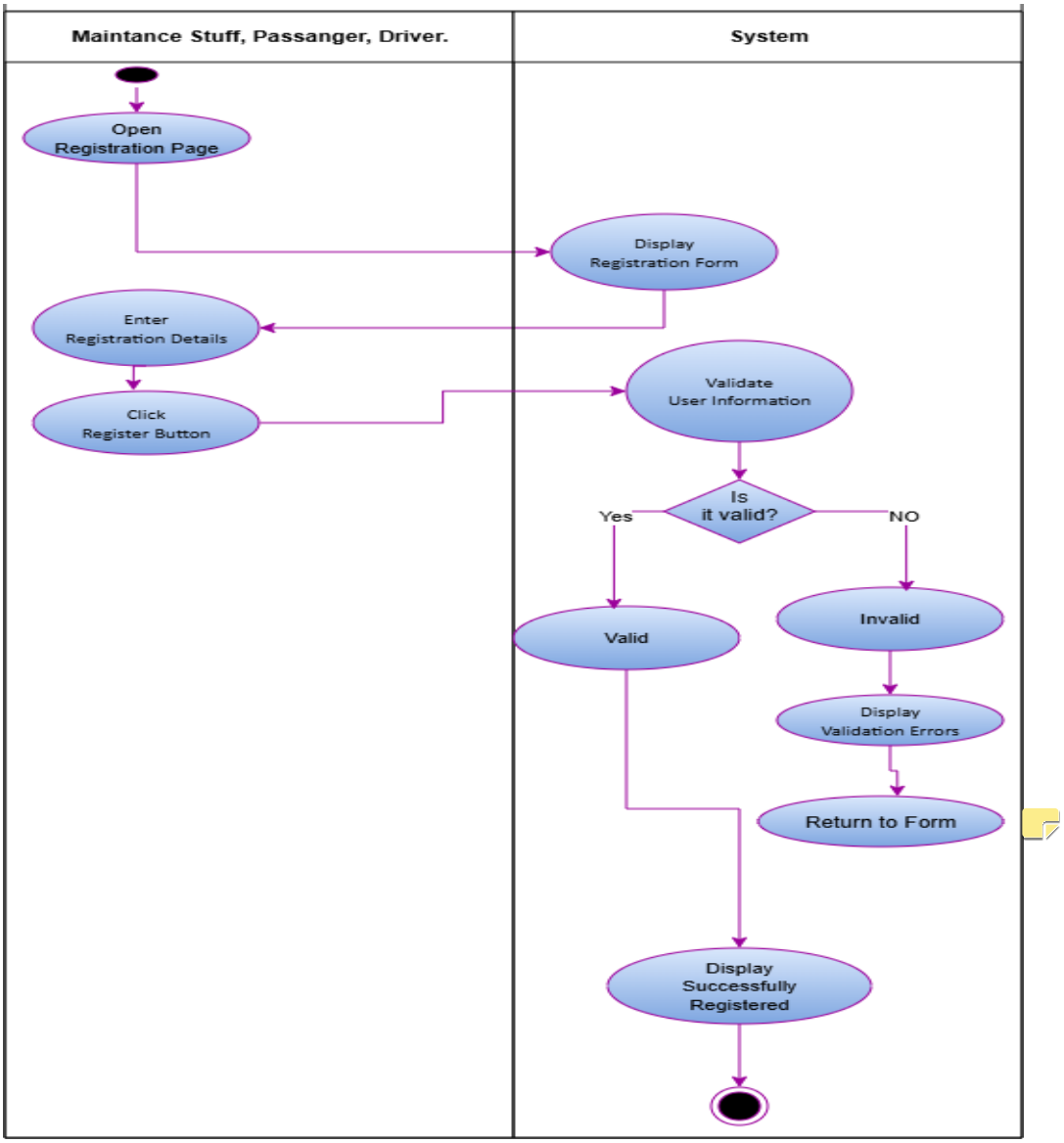


Figure 2: Register Activity Diagram

1.2.2. Login Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC2**		
User Case Name	**Login**		
Actor(s)	Fleet manager, Bus Driver, Passenger, Maintenance Staff		
Brief Description	The user enters their details (username and password) to access the system.		
Frequency of use	Daily, whenever a user wants to access the system.		
Preconditions	1. The user has a registered account. 2. The user is logged into the system using their email and password.		
Main Flow of Events	Actor	System	
	1. Clicks Login. 2. Enters email and password. 3. Click “Login”	1.1 Display login page. 3.1 Verify credentials 3.2 Grant access/Display error message and redirects back to input 3.3 End use case	
Alternative flow of events	1. User enters incorrect details. 2. The system shows an error message “Invalid login details”. 3. User tries again.		
Postconditions	The user has successfully logged into the system and taken to their dashboard.		
Special requirements	Passwords must be encrypted, login credentials must be validated, and access must be based on the user's role (Fleet manager, Maintenance staff, Driver, or Passenger).		
Created by	Yonwaba Mgqulwana	Last Modified by	Simanye Siximba
Date	30 July 2025	Date	06 August 2026

Table 3: Login Use Case Narrative

Activity Diagram 2: Login

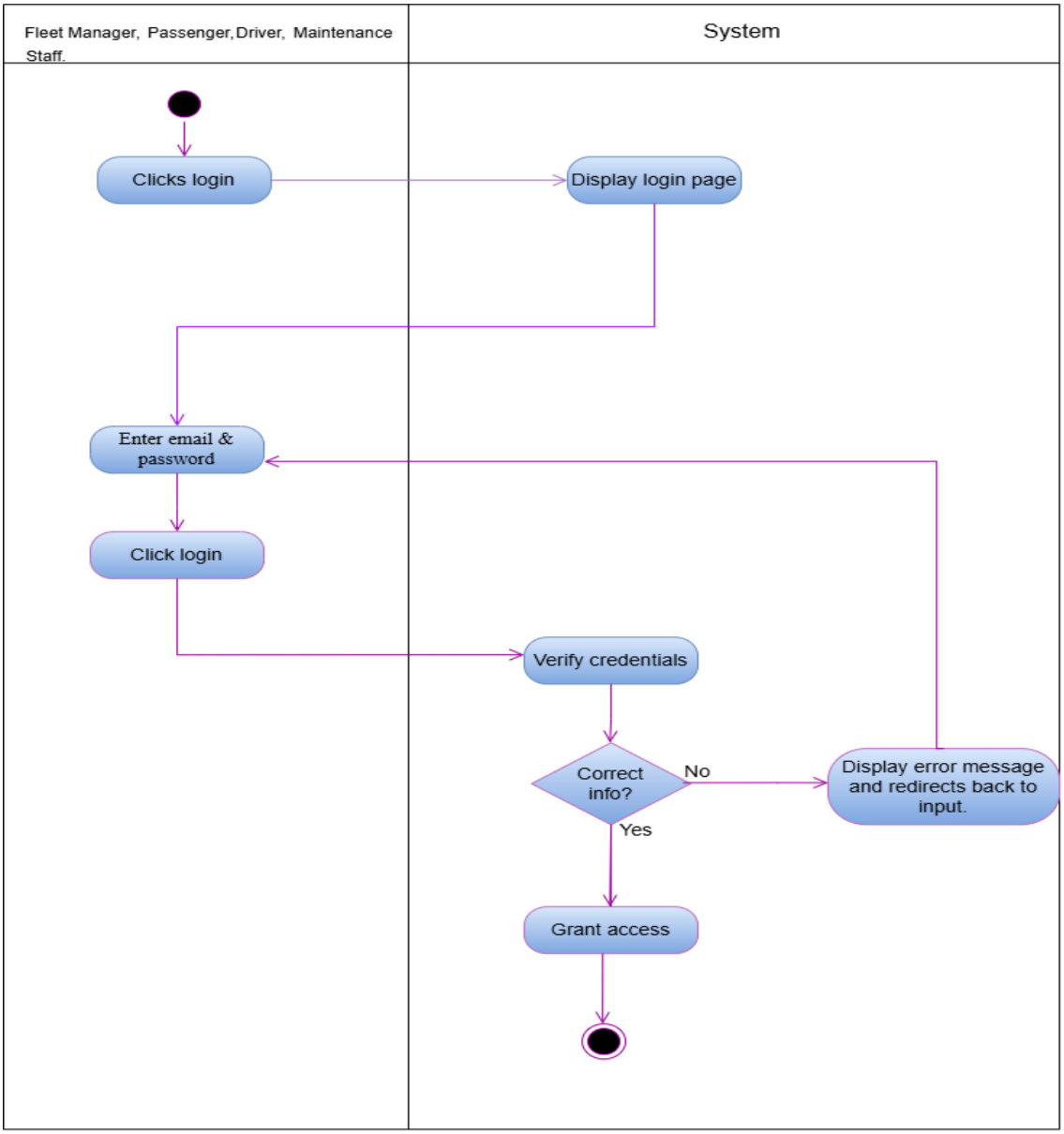


Figure 3: : Login Activity Diagram

1.2.3. Register Bus Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC3**		
User Case Name	**Register Buses**		
Actor(s)	Fleet Manger		
Brief Description	A fleet manager registers a new bus into the system by providing bus details such as registration number, model, manufacturer, seating capacity and fuel type. After successful registration, the bus becomes available for route assignment and scheduling.		
Frequency of use	High		
Preconditions	Fleet Manager has successfully logged into the system and has permission to register buses.		
Main Flow of Events	Actor	System	
	1. Selects “Register Buses”. 2. Enters the bus details (registration number, model, seating capacity, fuel type etc). 3. Clicks “Save/Register”.	1.1 Displays an empty bus registration form. 3.1 Validates required fields and checks registration number uniqueness. 3.2 Saves the bus record and displays a confirmation message. “Bus registered successfully”.	
Alternative Flows of Events	3.1.1 If required fields are missing or invalid, the system highlights the errors and remains on the form. 3.1.2 If the registration number already exists, the system reminds the fleet manager to enter a different registration number.		
Postconditions	New bus record is stored in the database, bus becomes available for route assignment, driver scheduling and maintenance scheduling.		
Special Requirements	Registration number must be unique; seating capacity must be a positive number.		
Created by	Anesipho Kalazana	Last Modified by	Elihle Nteyi
Date	03 August 2026	Date	07 August 2026

Table 4: Register Bus Use Case Narrative

Activity Diagram 3: Register Bus

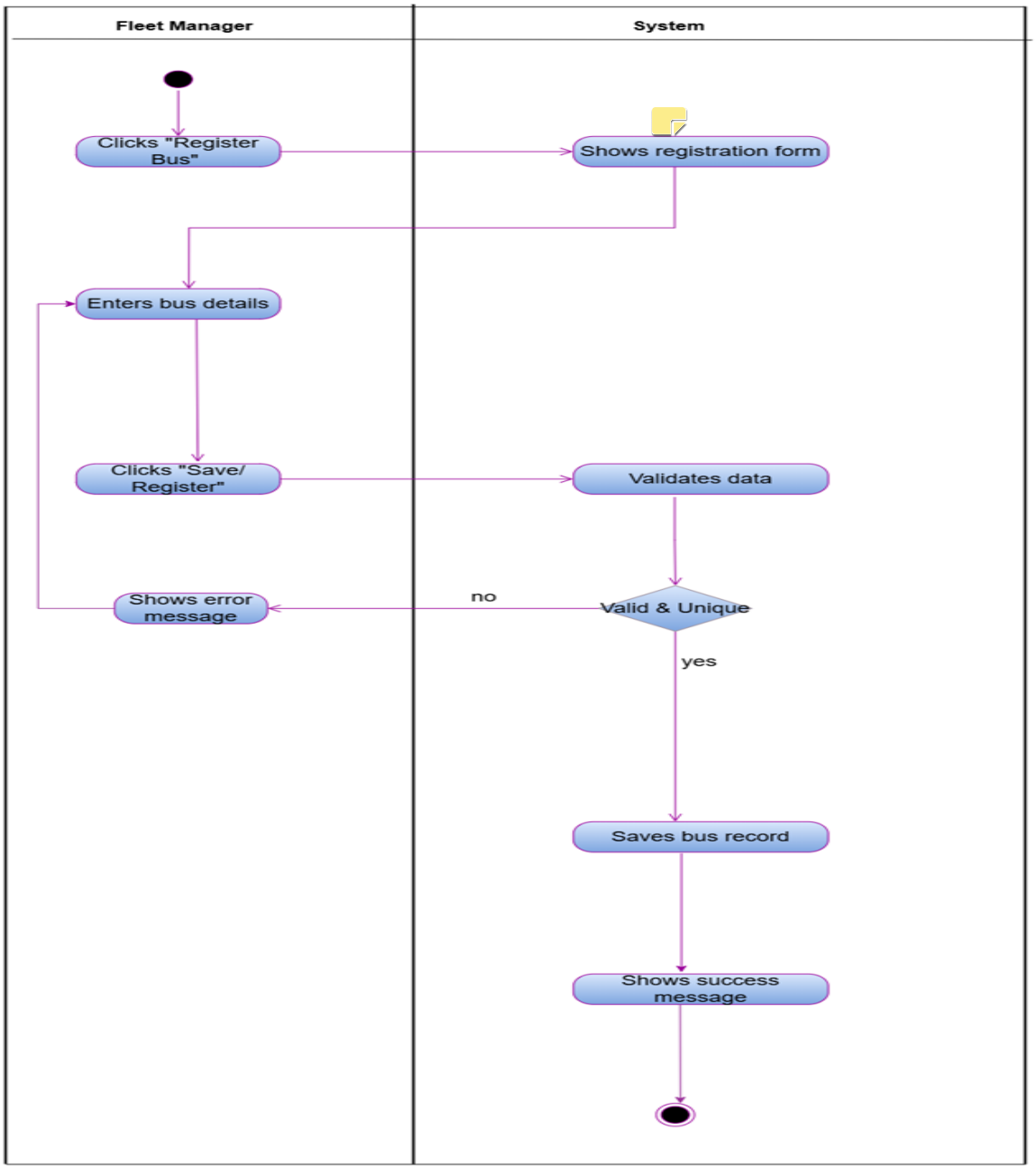


Figure 4: Register Bus Activity Diagram

1.2.4. Create Routes Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC4**		
User Case Name	**Create Routes**		
Actor(s)	Fleet Manager		
Brief Description	Describes how the Fleet Manager defines a new bus route in the system.		
Frequency of use	As needed, whenever new routes must be added or existing service areas expanded.		
Preconditions	The Fleet Manager is logged into the system with valid credentials.		
Main Flow of Events	Actor	System	
	1. Logs into the system. 2. Selects the "Create Route" option from the dashboard. 3. Enters the route details and adds intermediate stops. 4. Submits the route for saving	2.1. Displays the route creation form (route name/ID, origin, destination, stops, distance, estimated duration). 4.1. Validates the entered data (checks required fields and formats). 4.2. Checks that the route ID/name is unique. 4.3. Saves the new route to the database. 4.4. Displays a confirmation message to the Fleet Manager	
Alternative Flows of Events	4.1.1. If required fields are missing or incorrectly formatted, the system displays an error message and returns the Fleet Manager to step 3.		
Postconditions	Success: A new route record is created and saved in the system database.		
Special Requirements	Route names/IDs must be unique across the system.		
Created by	Nadia Mpohlekana	Last Modified by	Anesipho Kalazana
Date	28 July 2026	Date	07 August 2026

Table 5: Create Routes Use Case Narrative

Activity Diagram 4: Create Routes

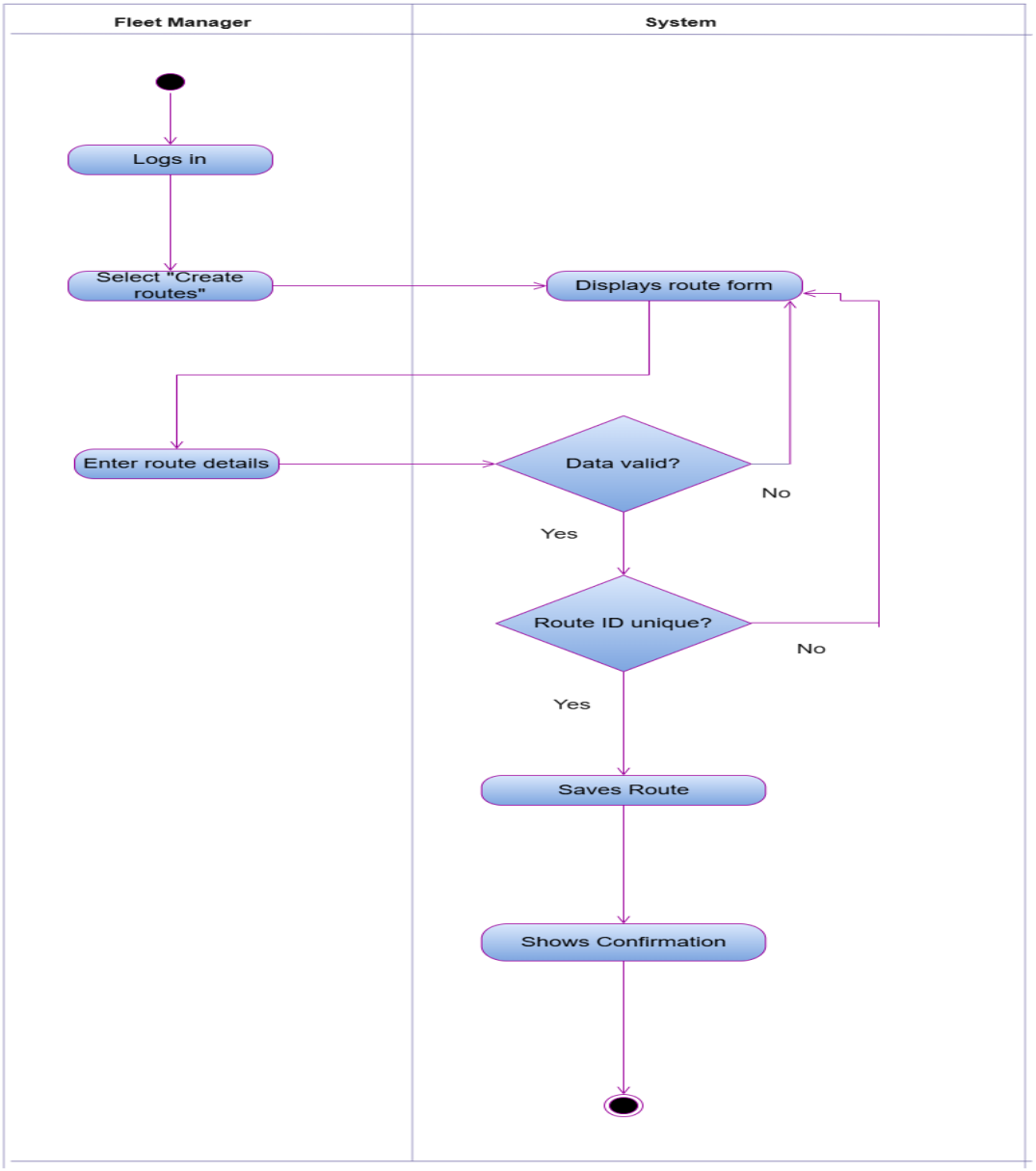


Figure 5: Create Routes Activity Diagrams

1.2.5. Assign Routes Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC5**		
User Case Name	**Assign routes**		
Actor(s)	Fleet manager		
Brief Description	The Fleet Manager assigns an available bus and driver to a specific route and schedule. The system validates the availability of the selected bus and driver before saving the assignment and updating the schedule		
Frequency of use	Whenever a new route needs to be assigned or an existing assignment needs to be updated		
Preconditions	The Fleet Manager must be logged into the system; driver must be valid before a route assignment can be performed		
Main Flow of Events	Actor	System	
	1. Logs in 2. Selects Assign Routes 3. Selects the route 4. Selects bus 5. Selects driver 6. Clicks Assign Route	1.1 Authenticates the Fleet Manager 2.1 Displays routes, buses, and drivers 3.1 Displays route details 4.1 Validates bus availability 4.2 Validates driver availability 4.3 Saves the route assignment 4.4 Updates the driver's schedule 6.1 Update bus allocation 6.2 Displays a confirmation message	
Alternative Flows of Events	6.1.1 If the selected bus unavailable the system shows an error message 6.2.1 If the selected driver is already assigned to another route, the system displays an error message		
Postconditions	Upon completion of the route assignment process, the system saves the assignment.		
Special Requirements	The system shall allow only authenticated Fleet Managers to assign routes by validating bus and driver's availability to prevent scheduling conflicts.		
Created by	Olebogeng Gumbu	Last Modified by	Esethu Mtose
Date	29 July 2026	Date	06 August 2026

Table 6: Assign Route Use Case Narrative

Activity Diagram 5: Assign Routes

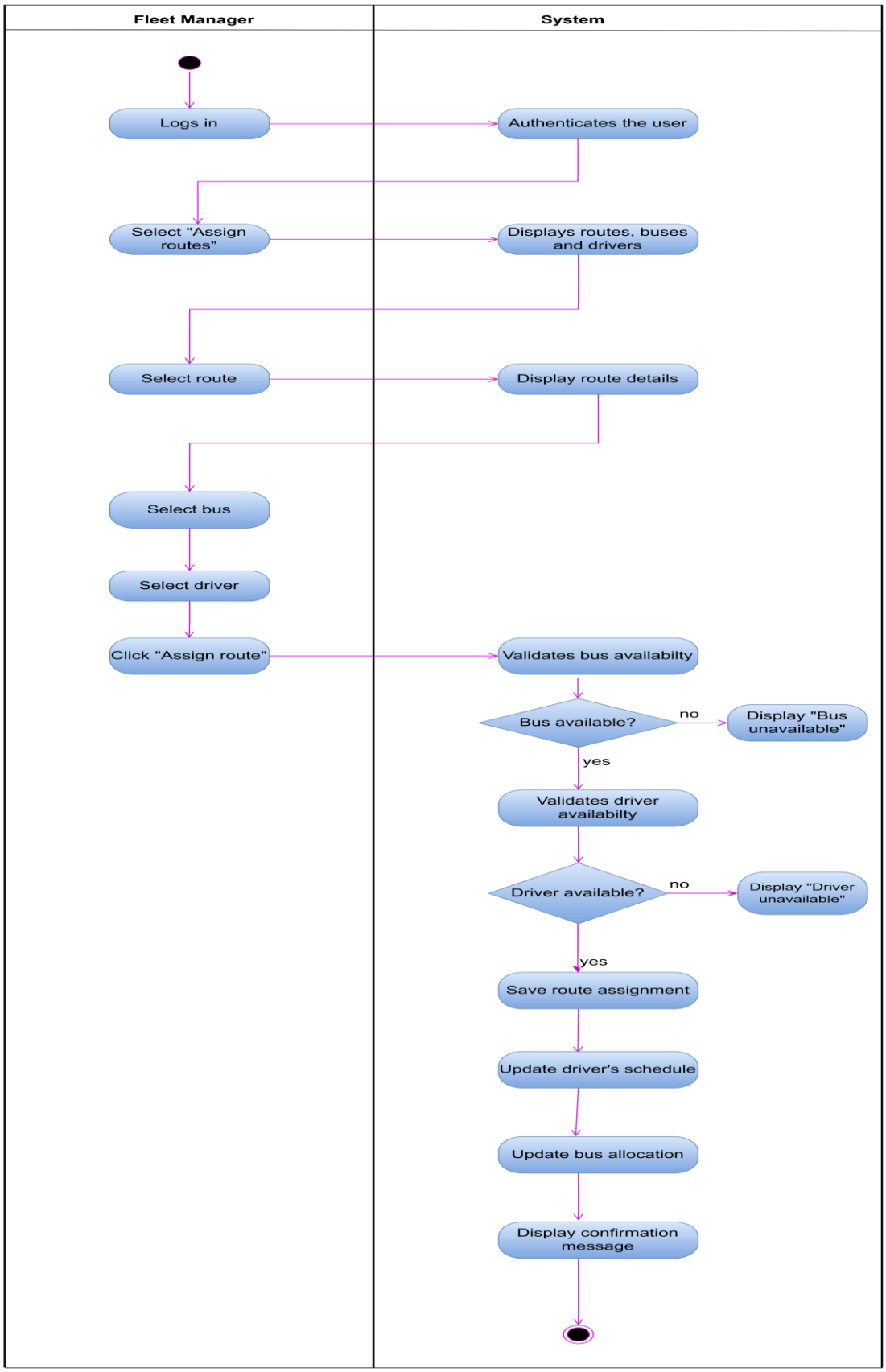


Figure 6: Assign Routes Activity Diagram

1.2.6. View Assigned Routes Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC6a**		
User Case Name	**View Assigned Routes**		
Actor(s)	Bus Driver		
Brief Description	The Bus Driver views the routes and trip schedule assigned to them, including trip details such as route path, stops, and pickup/drop-off times.		
Frequency of use	Daily		
Preconditions	1. The Bus Driver has a registered account and is logged into the system. 2. Routes/trips have been assigned to the driver.		
Main Flow of Events	Actor	System	
	1. Bus Driver logs in. 2. Bus Driver navigates to "My Schedule" / "Assigned Routes". 3. Bus Driver selects a route/trip to view further details.	1.1 Authenticate the user. 2.1 Retrieve and display the routes and trip schedule assigned to the driver. 3.1 Display detailed trip information (route path, stops, pickup/drop-off times, assigned bus).	
Alternative Flows of Events	1. No routes/trips have been assigned to the driver → System displays "No routes assigned".		
Postconditions	The Bus Driver has viewed their assigned routes and trip schedule. No data is modified by this use case.		
Special Requirements	Schedule information must reflect real-time updates; driver view must be usable on mobile devices.		
Created by	Siximba Simanye	Last Modified by	Yonwaba Mqulwana
Date	03 August 2026	Date	07 August 2026

Table 7: View Assigned Routes Use Case Narrative

Activity Diagram 6: Assigned Routes

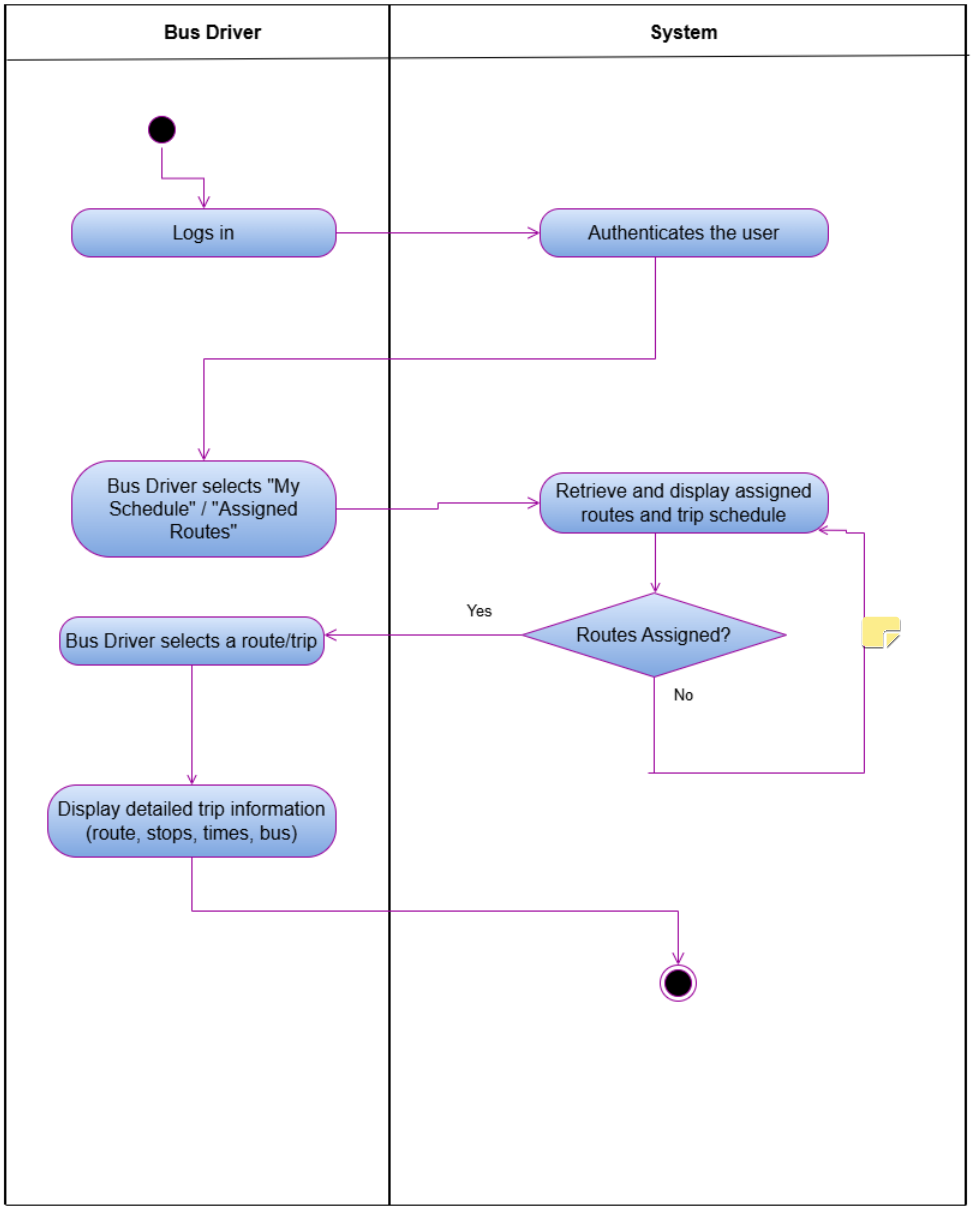


Figure 7: Assigned Routes Activity Diagram

1.2.7. View Maintenance Schedule Use Case Narrative

BusNova			
Use Case Specification			
Use Case Id	**UC6b**		
User Case Name	**View Maintenance Schedule**		
Actor(s)	Maintenance Staff		
Brief Description	Maintenance Staff view the maintenance and servicing schedule for buses in the fleet.		
Frequency of use	Medium		
Preconditions	1. Maintenance Staff has a registered account and is logged into the system.		
Main Flow of Events	Actor	System	
	1. Maintenance Staff logs in. 2. Maintenance Staff navigates to "Maintenance Schedule". 3. Maintenance Staff filters/searches records by bus, date, or status (optional). 4. Maintenance Staff selects a bus/service record to view further details.	1.1 Authenticate the user. 2.1 Display the Maintenance Schedule screen. 3.1 Retrieve and display the matching maintenance/service records. 4.1 Display detailed service information (last service date, next due date, service type, status). 4.2 If overdue, highlight the record and trigger an alert.	
Alternative Flows of Events	1. No maintenance records match the filter criteria → System displays "No maintenance records found".		
Postconditions	Maintenance Staff has reviewed the maintenance schedule, with overdue services flagged. No data is modified by this use case.		
Special Requirements	1. Schedule information must reflect real-time updates.		
Created by	Siximba Simanye	Last Modified by	Nadia Mpohlekana
Date	03 August 2026	Date	06 August 2026

Table 8: View Maintenance Schedule Use Case Narrative

Activity Diagram 7: View Maintenance Schedule

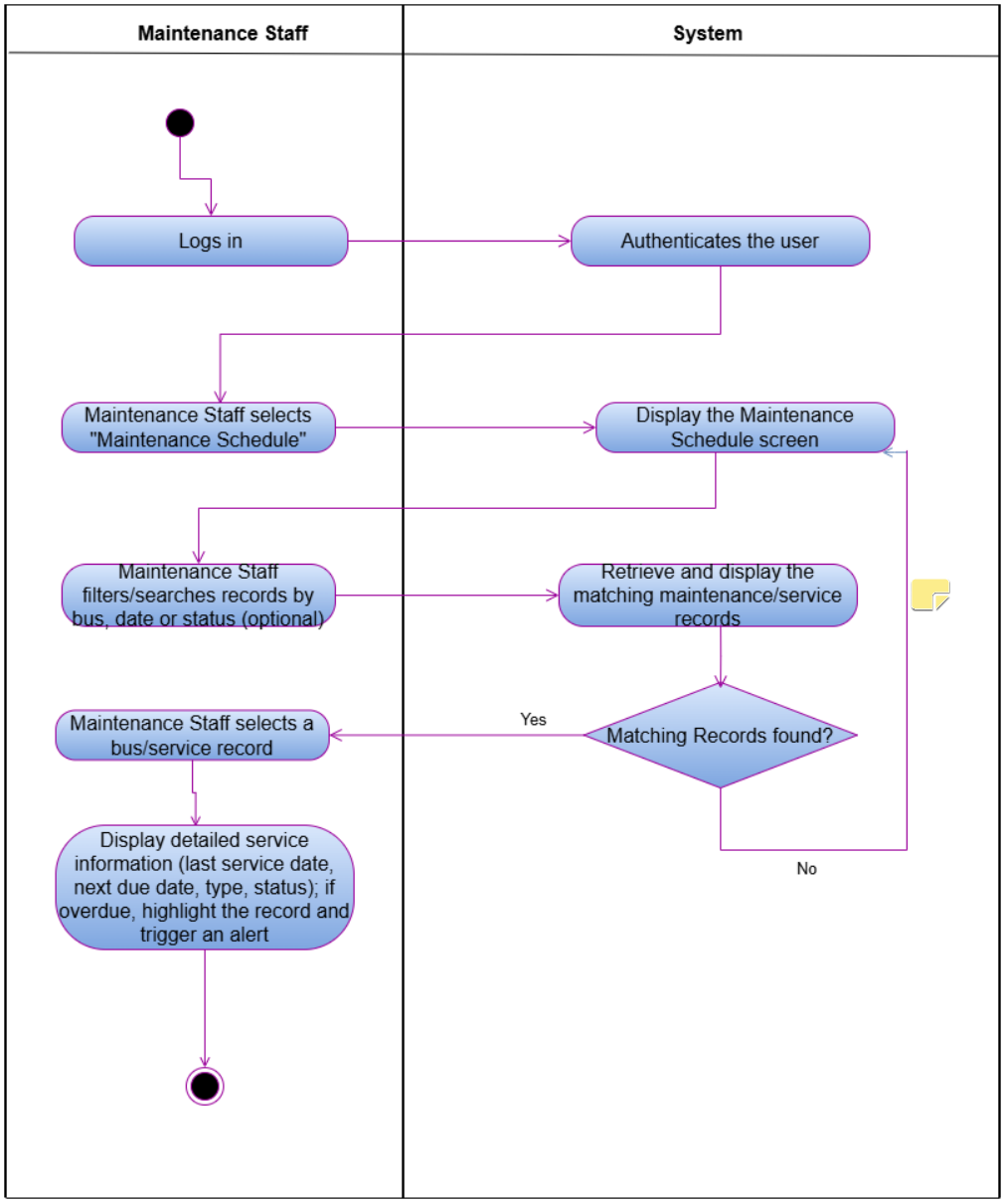


Figure 8: View Maintenance Schedule Activity Diagram

2.Section B: Data Modelling Requirements

2.1. Entity Relationship Diagram

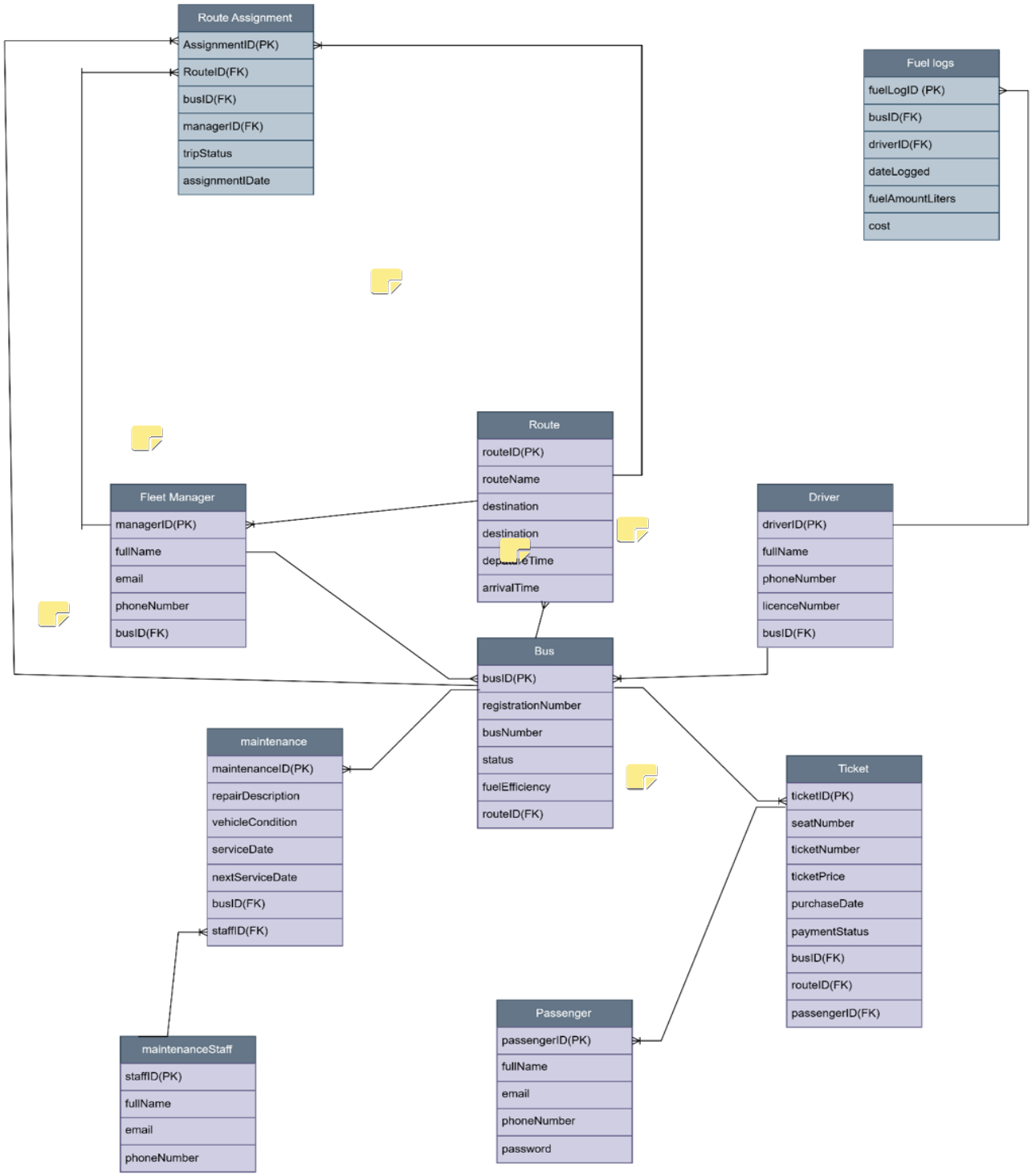


Figure 9: BusNova Entity Relationship Diagram

2. 2. Attribute Framing

2.2.1. Fleet Manager

Dia\LOCALDB... - dbo.Staff | Dia\LOCALDB...- dbo.Routes | Dia\LOCALDB...eAssignments | Dia\LOCALDB...o.Passengers | Dia\LOCALDB...tenanceStaff

Column Name	Data Type	Allow Nulls
StaffID	int	☐
Username	varchar(50)	☐
PasswordHash	varchar(255)	☐
FirstName	varchar(50)	☐
LastName	varchar(50)	☐
ContactNumber	varchar(15)	☐
Email	varchar(100)	☐
Role	varchar(20)	☐
DateHired	date	☐
IsActive	bit	☐
		☐

Figure 10: Fleet Manager

2.2.2. Passengers

Dia\LOCALDB....Passengers | Dia\LOCALDB...tenanceStaff | Dia\LOCALDB...ntenanceLogs | Dia\LOCALDB...dbo.FuelLogs | Dia\LOCALDB#... dbo.Drivers

Column Name	Data Type	Allow Nulls
PassengerID	int	☐
Username	varchar(50)	☑
PasswordHash	varchar(255)	☑
FirstName	varchar(50)	☐
LastName	varchar(50)	☐
ContactNumber	varchar(15)	☑
Email	varchar(100)	☑
		☐

Figure 11: Passengers

2.2.3. Drivers

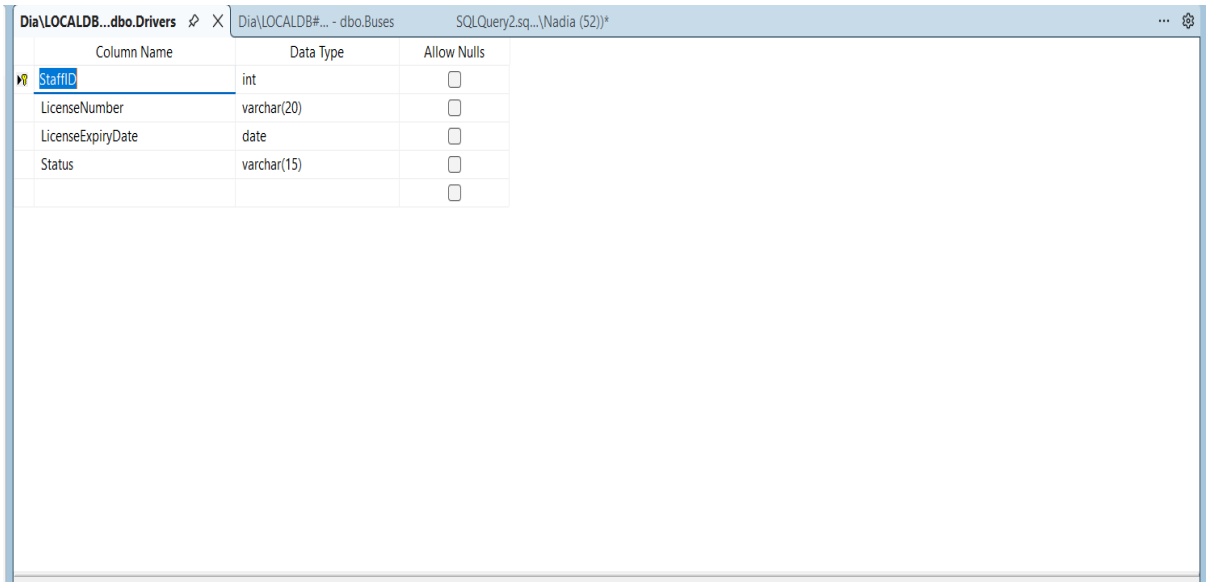


Figure 12: Drivers

2.2.4. Maintenance Staff

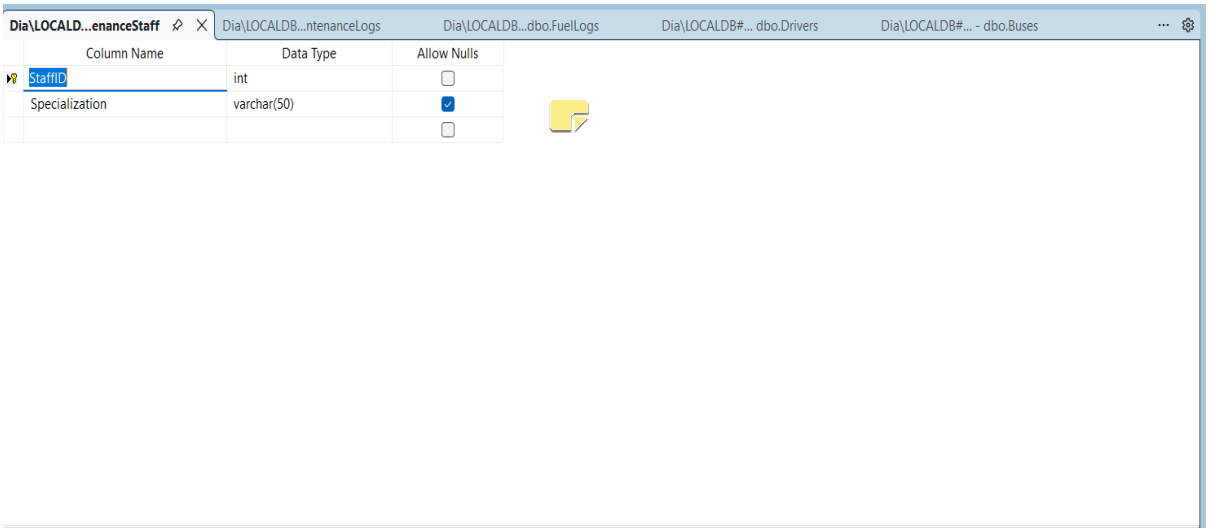


Figure 13: Maintenance Staff

2.2.5. Buses

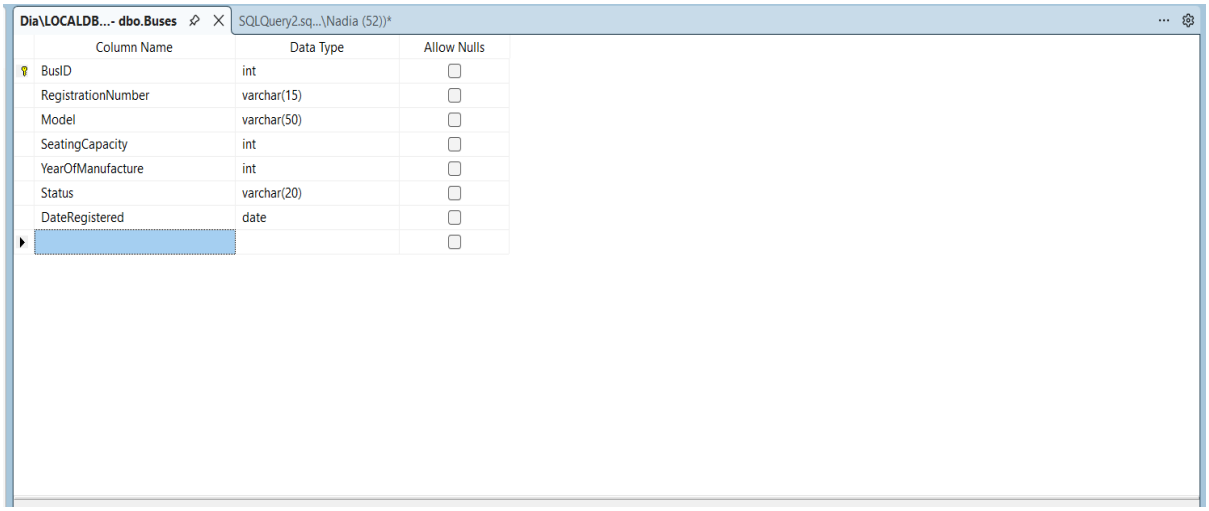


Figure 14: Buses

2.2.6. Routes

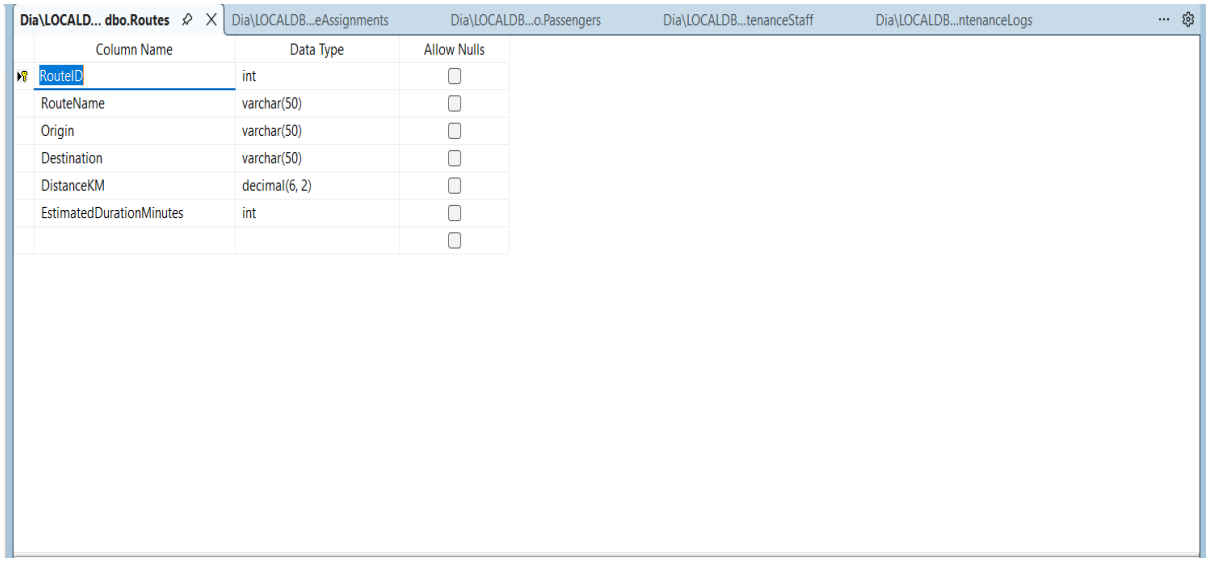


Figure 15: Routes

2.2.7. Route Assignments

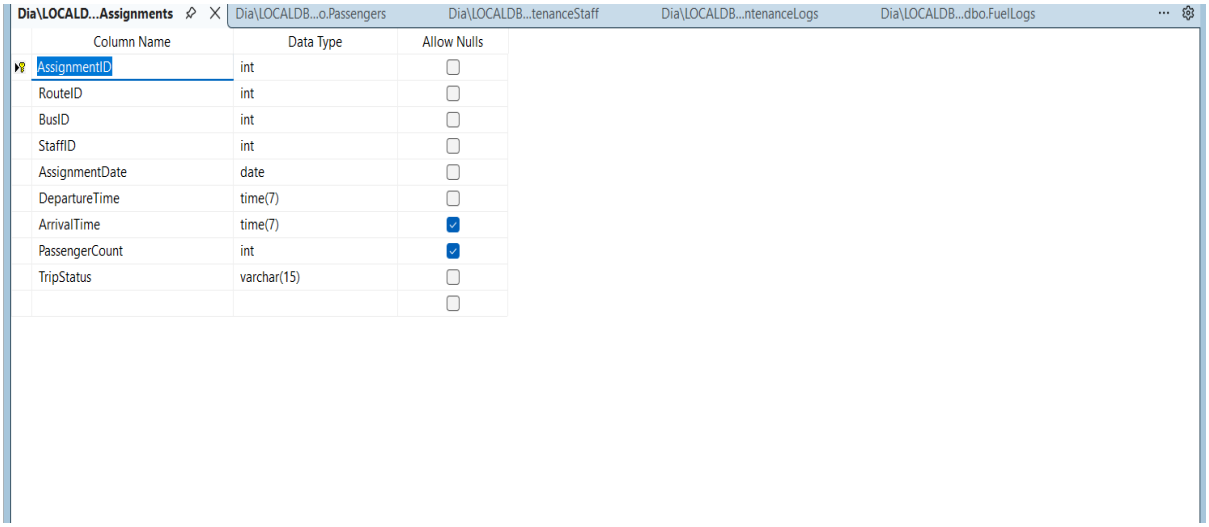


Figure 16: Route Assignments

2.2.8. Maintenance Logs

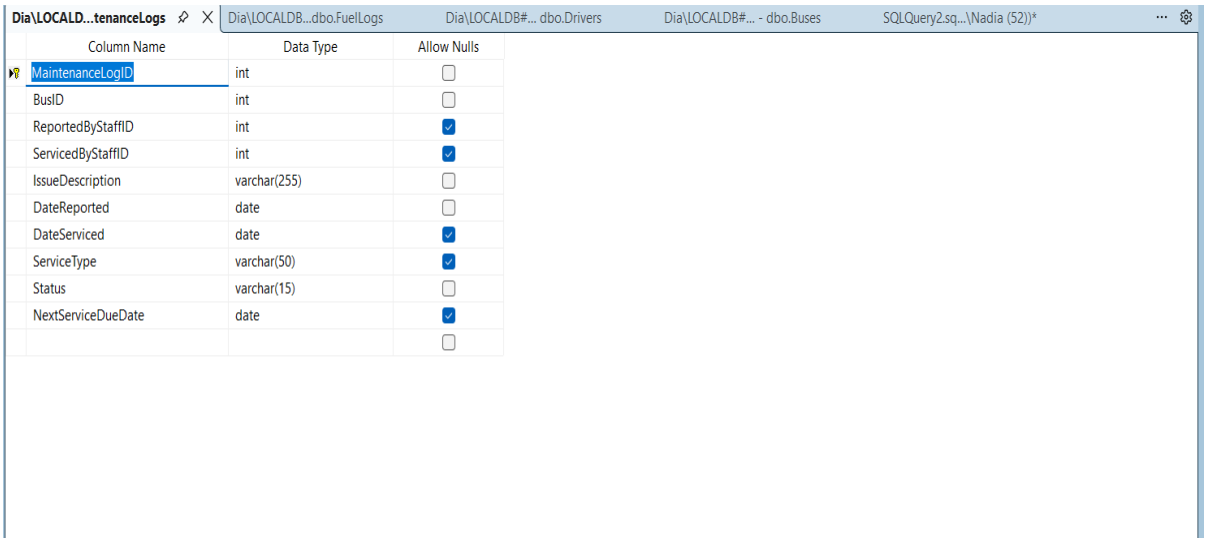


Figure 17: Maintenance Logs

2.2.9. Fuel Logs

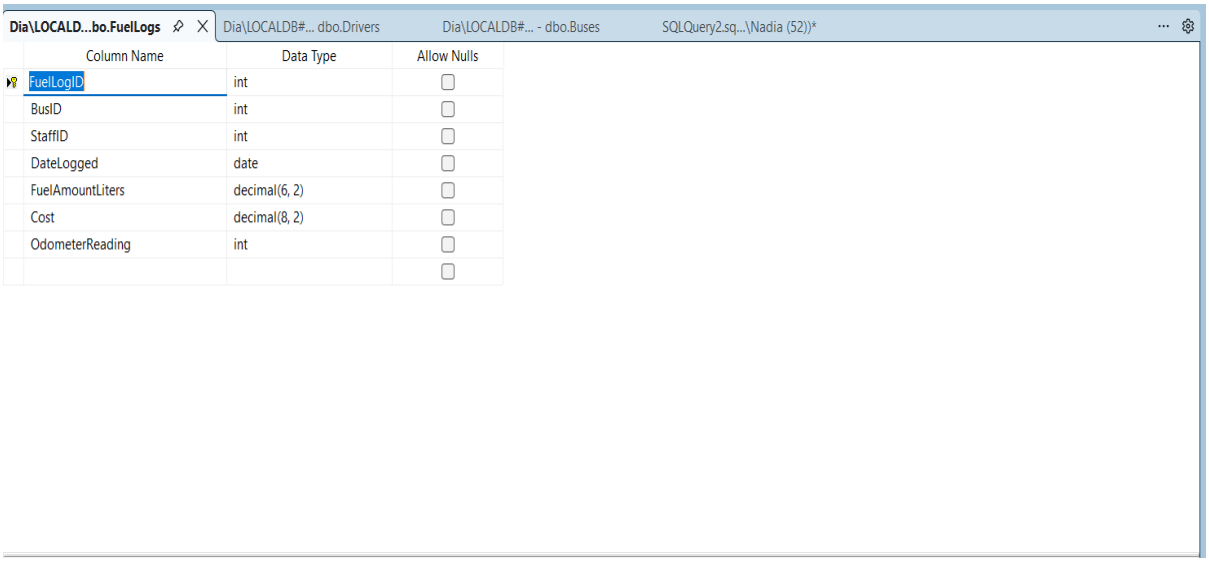


Figure 18: Fuel Logs

2.2.10. Tickets

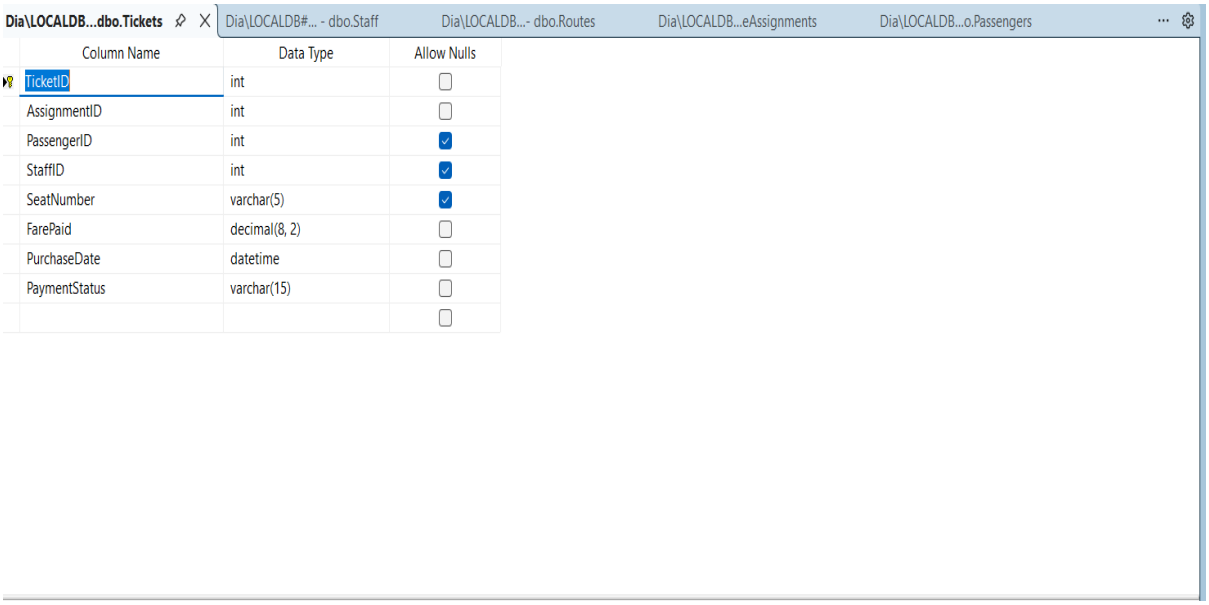


Figure 19: Tickets

3.Section C: Systems Design Requirements

3.1. Windows Navigation Diagram

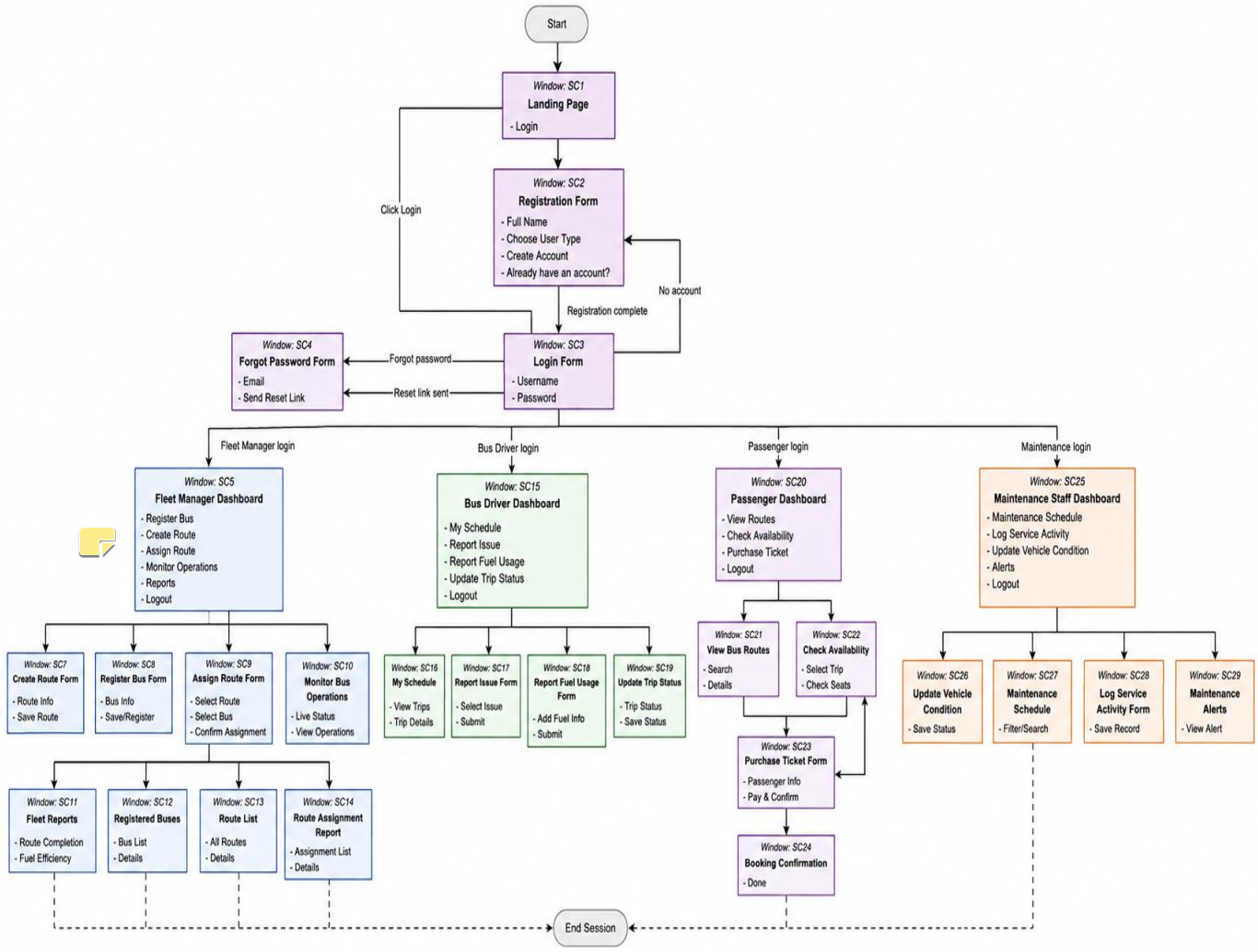


Figure 20: BusNova Windows Navigation Diagram

3.2. Screen Design Specification

3.2.1. Screen 01: Home Page

The Home Page serves as the main entry point to the BusNova System. It introduces users to the system and provides navigation options to access the relevant functionality. Users can select **Get Started**, **Login**, or **Register** to proceed to the appropriate part of the system.

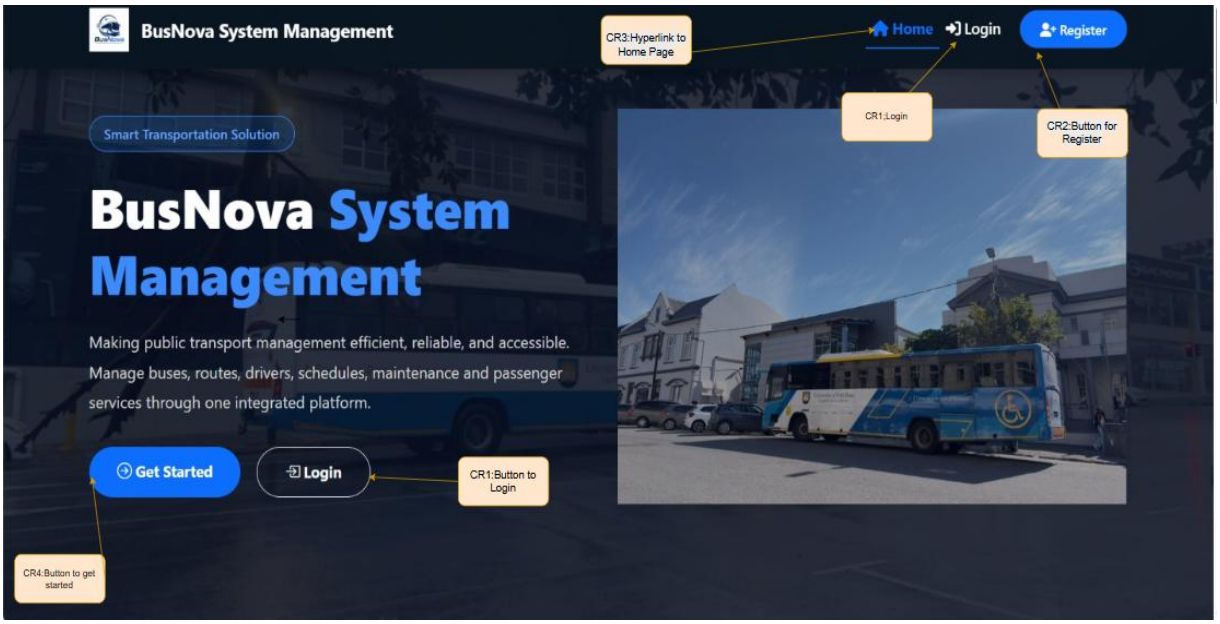


Figure 21: Home Page Screen

Screen Narrative and Controls: SC1

Reference	Control Type	Data Input	Data Output
CR1	Button	Click	Login page
CR2	Button	Click	Register page
CR3	Image		Bus image
CR4	Image		Background bus image

Table 9: SC1 - Screen Narrative and Controls

3.2.2. Screen 02: Register

The Register screen allows new users to create a BusNova account by entering their personal and account information, including their name, username, email address, contact number and password. The user also selects their appropriate system role during registration. The information provided is used to create the user's account and enable authenticated access to the system.

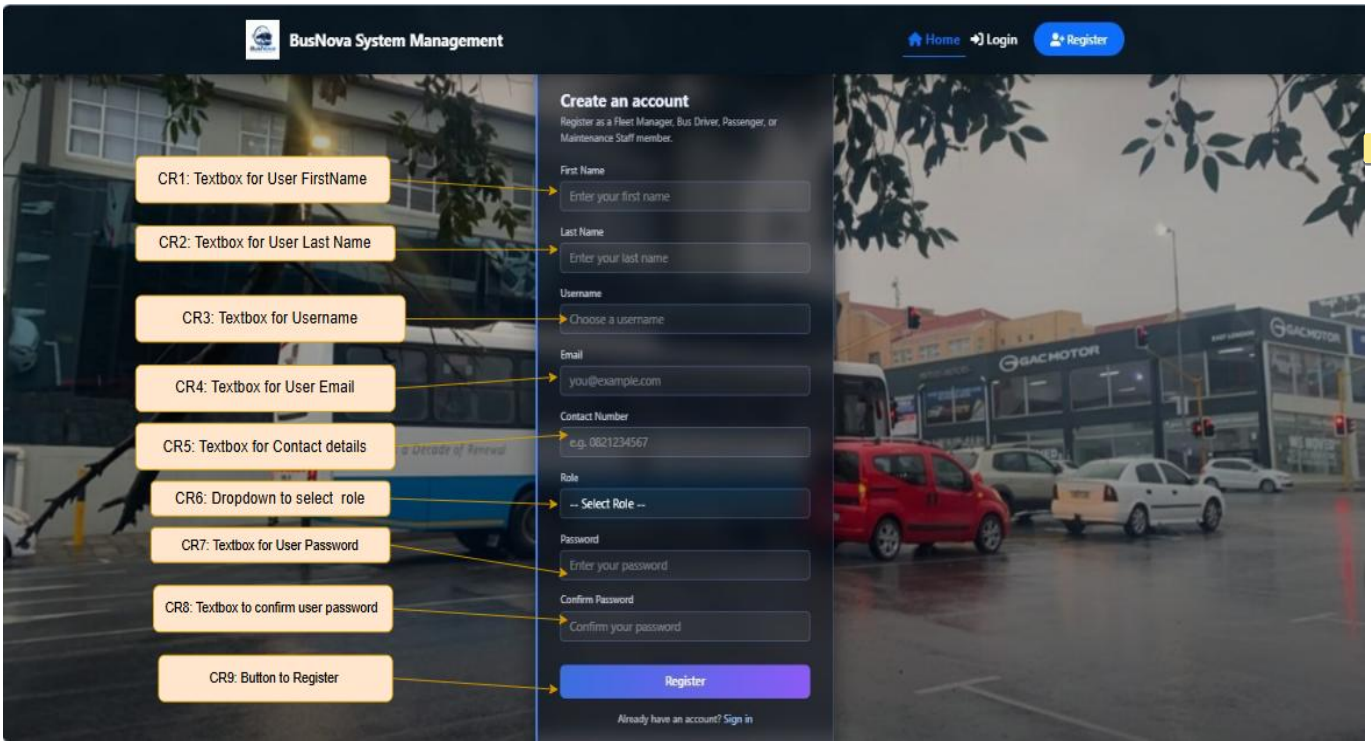


Figure 22: Register Screen

Screen Narrative and Controls: SC2

Reference	Control Type	Data Input	Data Output
CR1	Textbox	User's first name	
CR2	Textbox	User's last name	
CR3	Textbox	Username	
CR4	Textbox	User's email	
CR5	Textbox	Contact details	
CR6	Dropdown	Select a role	Role
CR7	Textbox	User enters password	
CR8	Textbox	Confirms user's password	
CR9	Button	Click	Login page

Table 10:SC2 - Screen Narrative and Controls

3.2.3. Screen 03: Login

The Login screen allows registered users to securely access the BusNova System by entering their email address and password. Upon successful authentication, the user is directed to the appropriate dashboard based on their assigned role.

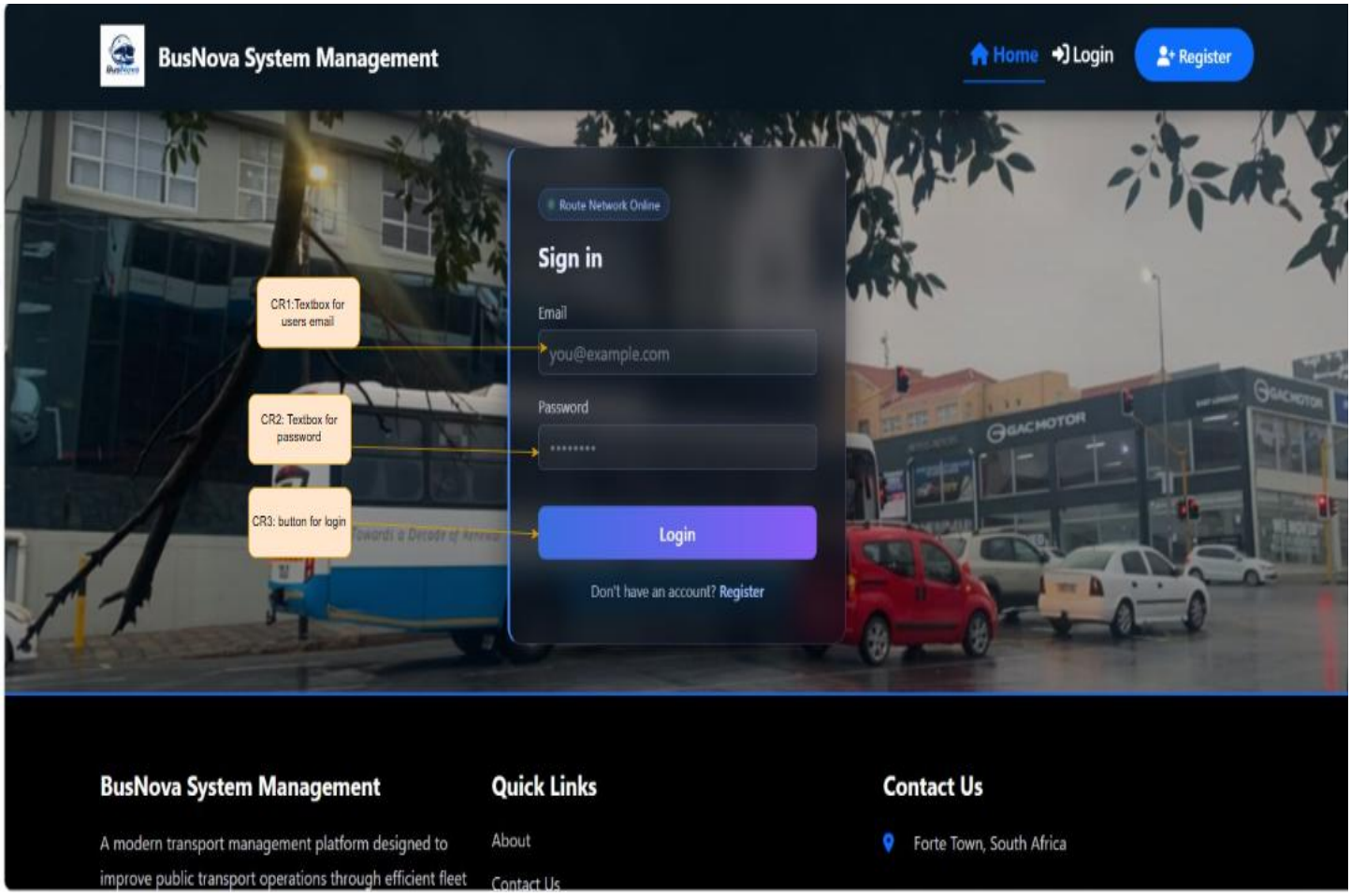


Figure 23: Login Screen

Screen Narrative and Controls: SC3

Reference	Control Type	Data Input	Data Output
CR1	Textbox	User's email	
CR2	Textbox	User enters password	
CR3	Button	Click	Display dashboard

Table 11: SC3 - Screen Narrative and Controls

3.2.4. Screen 04: Register Bus

The Register Bus screen enables the Fleet Manager to add a new bus to the BusNova fleet by capturing details such as the registration number, model, manufacturer, seating capacity, fuel type and year of manufacture. The **Register Bus** button saves the bus information to the system.

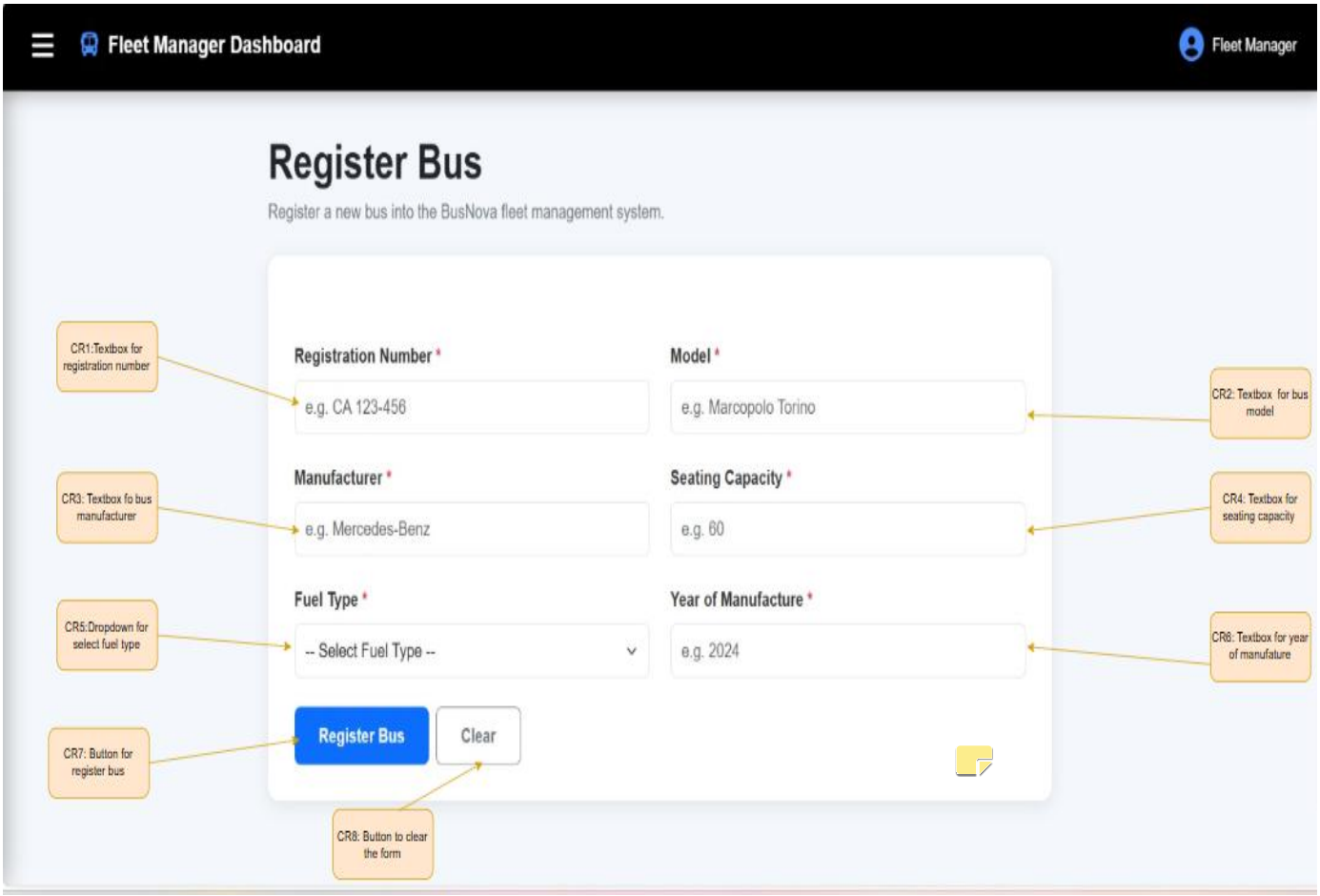


Figure 24: Register Bus Screen

Screen Narrative and Controls: SC4

Reference	Control Type	Data Input	Data Output
CR1	Textbox	Registration number	
CR2	Textbox	Bus model	
CR3	Textbox	Bus manufacturer	
CR4	Textbox	Seating capacity	
CR5	Dropdown	Select fuel type	Fuel type
CR6	Textbox	Year of bus manufacture	
CR7	Button	Click	Register bus
CR8	Button	Click	Clear the form

Table 12: SC4 - Screen Narrative and Controls

3.2.5. Screen 05: Create Routes

The Create Route screen enables the Fleet Manager to create a new bus route by entering the required route information, including the route name, origin, destination, distance and estimated travel time. The entered information is submitted to create and store the new route in the system.

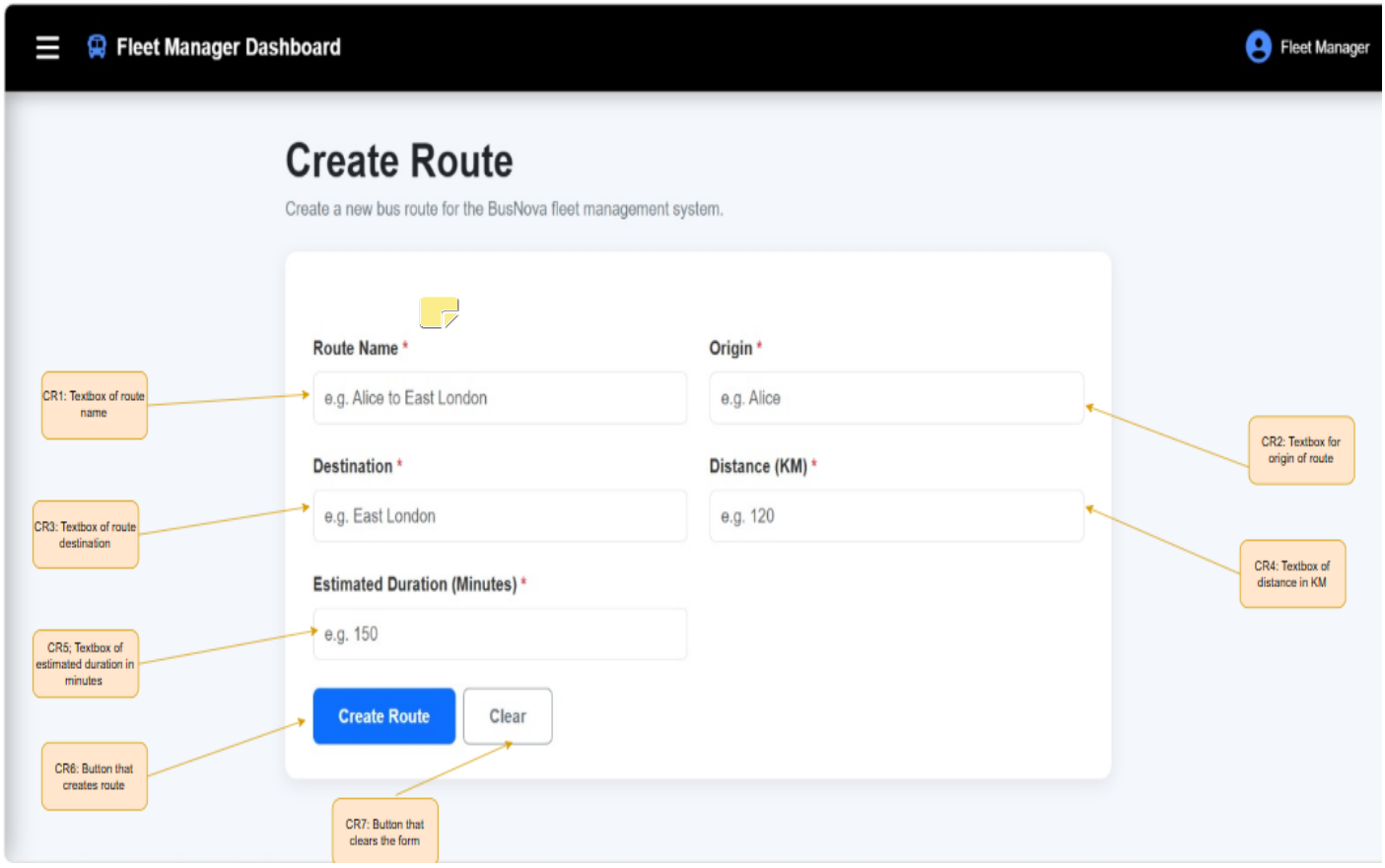


Figure 25: Create Routes Screen

Screen Narrative and Controls: SC5

Reference	Control Type	Data Input	Data Output
CR1	Textbox	Route name	
CR2	Textbox	Origin of route	
CR3	Textbox	Route destination	
CR4	Textbox	Distance in KM	
CR5	Textbox	Estimated duration in minutes	
CR6	Button	Click	Creates route
CR7	Button	Click	Clear the form

Table 13: SC5 - Screen Narrative and Controls

3.2.6. Screen 06: Assign Routes

The Assign Routes screen enables the Fleet Manager to assign an available bus and driver to a selected route and schedule. The Fleet Manager provides the required assignment details, including the route, driver, bus, date and departure information, allowing the system to allocate resources to a specific trip.

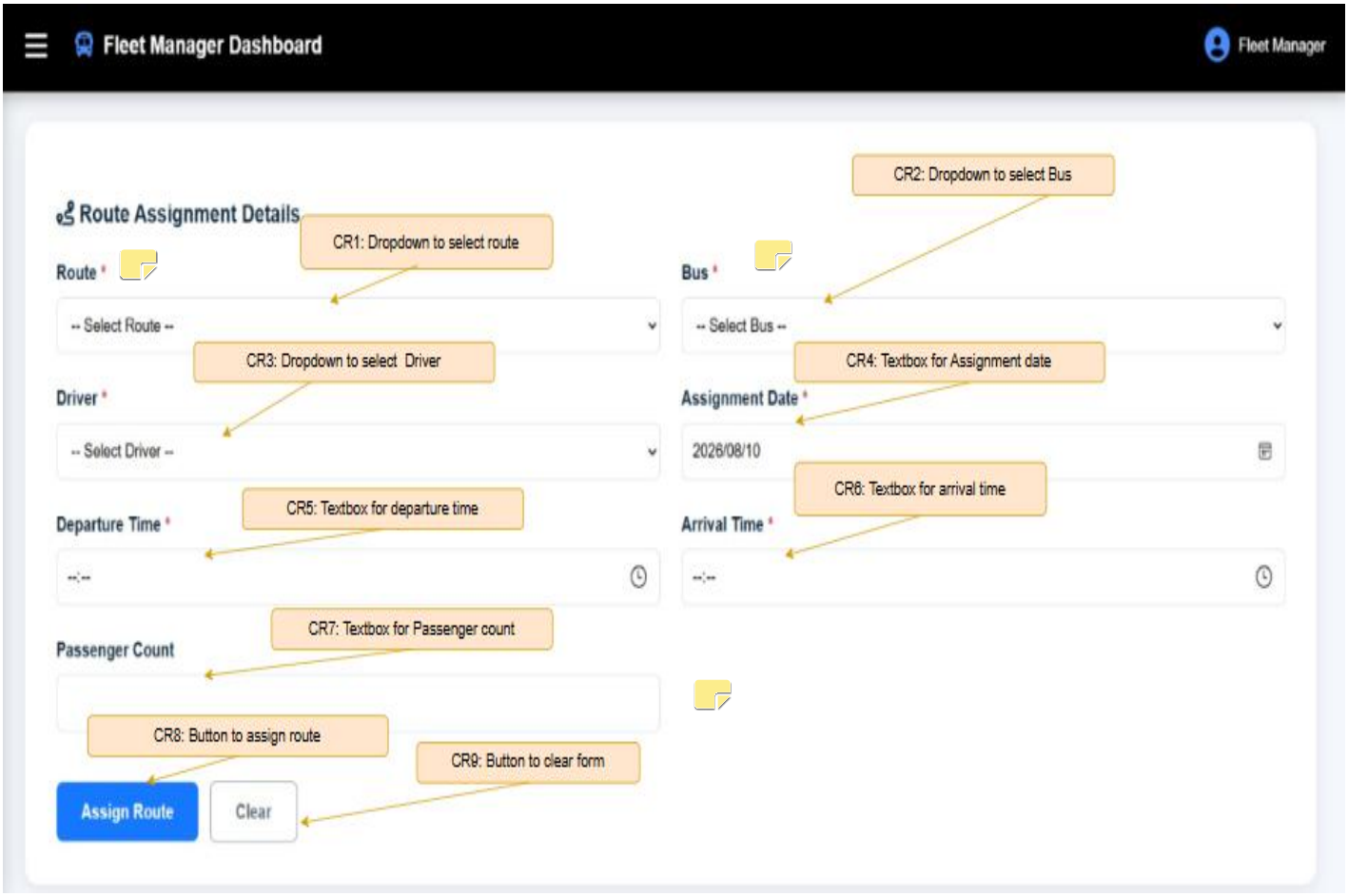


Figure 26: Assign Routes Screen

Screen Narrative and Controls: SC6

Reference	Control Type	Data Input	Data Output
CR1	Dropdown	Select route	Route
CR2	Dropdown	Select a bus	Bus
CR3	Dropdown	Select a driver	Driver
CR4	Textbox	Assignment date	
CR5	Textbox	Departure time	
CR6	Textbox	Arrival time	
CR7	Textbox	Passenger count	
CR8	Button	Click	Assign route
CR9	Button	Click	Clear the form

Table 14: SC6 - Screen Narrative and Controls

3.2.7. Screen 07: View Assigned Routes

The View Assigned Routes screen allows the Bus Driver to access their assigned routes and trip schedules. It displays important trip information, including the assigned route, bus, date, departure and arrival times, passenger count and current trip status, enabling the driver to monitor their scheduled transportation duties.

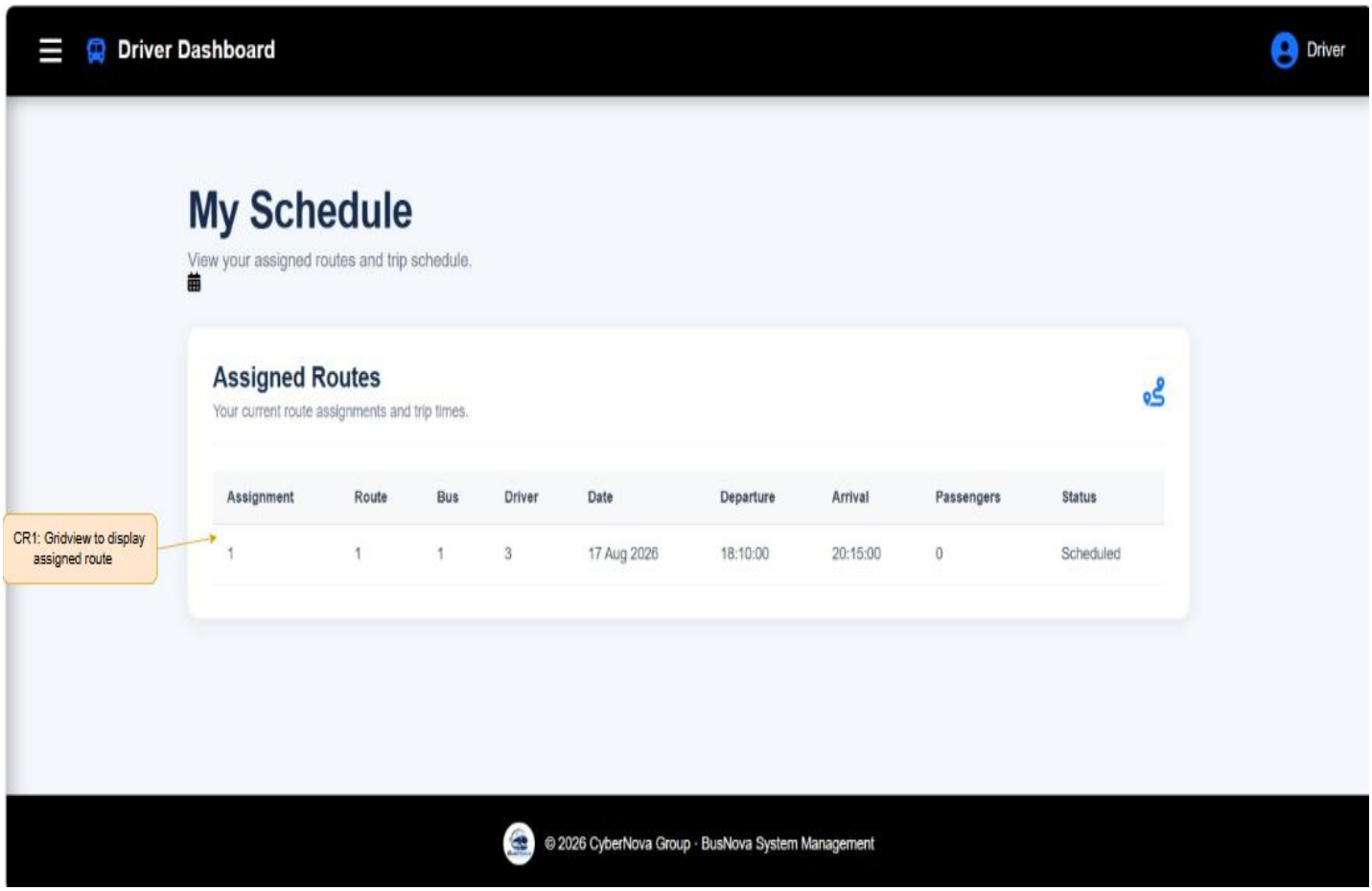


Figure 27: View Assigned Routes Screen

Screen Narrative and Controls: SC7

Reference	Control Type	Data Input	Data Output
CR1	Grid view	Display assigned route	

Table 15: SC7 - Screen Narrative and Controls

Sprint Backlog - Sprint 1

Requirement	Sub Tasks	Dependency	Estimated Completion Time (in days)	Responsible Group Member
1.Business Process Description	1.1 Identify actors and business processes	none	1 day	Nadia Mpohlekana
	1.2 Create System Matrix	Identify actors and business processes	1 day	Olebogeng Gumbu
	1.3 Develop Use Cases	Create System Matrix	2 days	Esethu Mtose
	1.4 Write Use Narratives	Developed use cases	2 days	Anesipho Kalazana
	1.5 Create Activity Diagrams	Use Case Narratives	2 days	Yonwaba Mqulwana
2.Data Modelling	2.1Identify entities and attributes	System matrix	1 day	Elihle Nteyi
	2.2 Create ERD	Entity identification	2 days	Esethu Mtose
	2. 3 Create Attribute Framing	ERD complete	1 day	Olebogeng Gumbu
	2.4 Generate SQL database creation script	Attribute framing complete	2days	Nadia Mpohlekana
3.System Design requirements	3.1UML Window Navigation Diagram	Activity Diagrams completed	2 days	Elihle Nteyi

	3.2 Screenshots	UML Window Navigation Diagram	1 day	Simanye Siximba
	3.3 Screen narratives and controls	System Screens (Screenshots) completed	1 day	Simanye Siximba
4.Documentation	Review sprint 1 document	All tasks completed	3 days	All members

Table 16: Sprint Backlog

Risk Log

Ref No	Description	SQERT	Probability	Impact	Criticality	Proximity	Author	Mitigation and owner	Date Raised	Date Updated	Current Status	Date Closed
1	Database changes made by one team member causing compatibility problems for other team members	ST	3	4	12	During collaborative development	Team Members	DB Developer	31/07/2026	08/08/2026	Reducing	Not closed
2	Insufficient system testing before submission	T	3	5	15	During final testing	Team Members	Testing Member	01/08/2026	08/08/2026	New	Not closed
3	Loss or corruption of project files or database due to inadequate backups	QER	2	5	10	Throughout development	Team Members	Technical Lead	02/08/2026	08/08/2026	Reducing	Not closed

Table 17: Risk Log

KEY: SQUERT

Code	Meaning
S	Scope
Q	Quality
E	Effort
R	Resources
T	Time

Sprint Progress Report

Project Name: BusNova Bus Management System	**Sprint Number:** Sprint 1
Sprint Duration: 13 July 2026 - 11 August 2026	**Report Date:** 11 August 2026
Report Number: 1	**Prepared By:** Nadia Mpohlekana

Sprint Goals	Goal 1: Analyse the business processes of the BusNova system and finalise the functional and non-functional requirements. Goal 2: Produce a complete UML design blueprint for the system, including use case, activity, and structural diagrams. Goal 3: Establish the technical foundation of the system by implementing the SQL Server database and an initial connected user interface.
Completed Work in line with Sprint Backlog	Task 1: Requirements and System Analysis - Completed Task 2: UML and System Design - Completed Task 3: Database and User Interface Foundation - Completed
In-Progress Work	Task 1: Implementation of the remaining system modules using the completed UML designs. Task 2: Development of business logic for user authentication, route management, booking processes, and administrative functions. Task 3: Testing database connectivity, validating user interface functionality, and refining screen layouts.
Upcoming Work	Task 1: Implement the remaining CRUD functionality for all system entities. Task 2: Conduct integration testing between the user interface and the SQL Server database. Task 3: Prepare the system for Sprint 2 by implementing advanced functionality, improving usability, and resolving identified issues.
Blockers and Challenges	Blocker 1: Coordinating multiple documentation deliverables simultaneously required careful planning and communication among team members. Blocker 2: Ensuring consistency across all UML diagrams and maintaining alignment with the business processes required several collaborative review sessions. Blocker 3: Integrating the initial user interface with the SQL Server database presented minor configuration challenges, which were resolved through collaborative troubleshooting and testing.

Team Member Contributions	Nadia Mpohlekana - Responsible for the completion, structure, and quality of the Sprint Progress Report. Yonwaba Mqulwana- Developed the Activity Diagrams and the System Matrix for the entire system. Anesipho Kalazana- Authored the Use Case Narratives. Esethu Mtose- Developed the Entity Relationship Diagram (ERD) and the Use Case Diagram. Elihle Nteyi- Generated the Database script and developed the Windows Navigation Diagram. Olebogeng Gumbu - Identified each system screen, detailing its contents and framing the required attributes. Simanye Siximba- Built the user interface and implemented the database.
Velocity and Burndown Chart	Velocity: The team achieved approximately 75% of the planned Sprint 1 deliverables. All planned design artefacts were completed successfully, with implementation activities commencing as scheduled.
Sprint Retrospective Summary	What Went Well: All planned UML and design deliverables were completed on schedule. What Didn't Go Well: Coordinating parallel documentation work across multiple team members occasionally caused version and alignment issues. Action Items for Improvement: Introduce a shared review checkpoint before finalising diagrams to catch inconsistencies earlier.
Key Takeaways	Sprint 1 successfully established the full analysis, design, and technical foundation of the BusNova Bus Management System. With requirements confirmed, UML artefacts completed, and the database and UI foundation in place, the team is well positioned to move into full module implementation and integration testing in Sprint 2.
Prepared by	Nadia Mpohlekana
Role	Project Manager & Documentation Lead
Date	11 August 2026

Table 18: Sprint Progress Report